Three-dimensional kinetic analysis of side-foot and instep soccer kicks

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ABSTRACT

NUNOME, H., T. ASAI, Y. IKEGAMI, and S. SAKURAI. A Three-dimensional kinetic analysis of side-foot and instep soccer kicks. *Med. Sci. Sports Exerc.*, Vol. 34, No. 12, pp. 2028–2036, 2002. **Purpose:** The purpose of this study was to identify the kinetic aspects of side-foot and instep soccer kicks to understand the different mechanics underlying the two kicks. **Methods:** The motions of both kicks were captured using a three-dimensional cinematographic technique. The kicking leg was modeled as a three-link kinetic chain composed of thigh, shank, and foot, from which joint torques and angular velocities were computed. **Results:** The ball velocity of the side-foot kick (23.4 ± 1.7 $m \cdot s^{-1}$) was significantly slower than that of the instep kick (28.0 ± 2.1 $m \cdot s^{-1}$). Significant differences were also observed between the two kicks for the magnitude of hip external rotation torque (56 ± 12 N·m in the side-foot kick; 33 ± 8 N·m in the instep kick) and hip external rotation angular velocity (11.1 ± 2.4 $rad \cdot s^{-1}$ in the side-foot kick; 6.0 ± 2.0 $rad \cdot s^{-1}$ in the instep kick). **Conclusion:** These results indicated that to hit the ball with the medial side of the foot, a complicated series of rotational motions are required for the side-foot kick. The hip external rotation torque dominantly exhibited in the side-foot kick caused the clockwise rotation of the thigh-shank plane at the later stage of kicking. This may allow the hip external rotation motion to increase directly the forward velocity of the side foot, with which players can squarely impact the ball. **Key Words:** KICKING MECHANICS, DIRECT LINEAR TRANSFORMATION, SIDE-FOOT SOCCER KICK, INSTEP SOCCER KICK

A wide range of skills form the foundation of soccer performance, but only one has been the subject of detailed biomechanical analysis. Kicking is the most widely studied skill in soccer (8). Among many forms of kicking, players use the most appropriate one depending on its nature and intent. In general, the side-foot kick is the most frequently used technique when a shorter and precise pass or shot is required, whereas the instep kick is used when a faster ball speed must be generated. According to a study that analyzed the techniques used to score the goals in the last World Cup (6), these two kicks were most commonly used to score goals. In particular for the side-foot kick, this technique was used to score 16 of 17 goals from penalty kicks.

Although several studies have examined the soccer kick (3,10–13,17), most were done on the quasi-planar aspects of the instep kick or similar maximal soccer kicks. Thus, minimal information is available especially for the side-foot kick. Recently, Levanon and Dapena (9) compared the three-dimensional angular motions required in the side-foot kick with those required in the instep kick. However, no study has focused on the three-dimensional kinetics of these kicking techniques. Thus, the true kinetics of the two kicks have not been quantified appropriately.

For understanding how the leg swing is produced in both kicks, it is necessary to identify its three-dimensional kinetic aspects. The purpose of this study, therefore, was to investigate the kinetic aspects of the two soccer kicks, side-foot and instep kicks, using a three-dimensional video analysis technique.

METHODS

Five experienced male high-school soccer players (height = 174.6 ± 4.9 cm; mass = 67.6 ± 4.8 kg) volunteered to participate in this study. Permission for participation was obtained from the participants and from the Human Research Committee of the Research Center of Health, Physical Fitness and Sports of Nagoya University. Informed written consent was obtained from each participant. Each participant preferred to kick the ball using the right leg.

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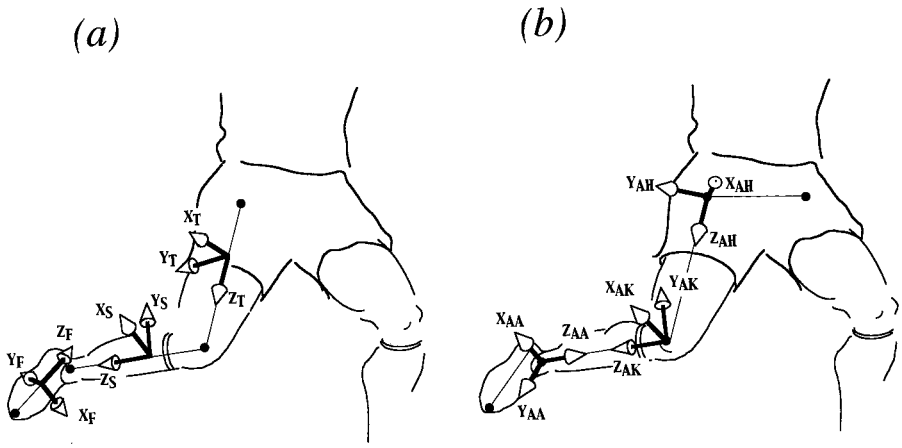

FIGURE 1—Reference frames. The reference frames (a) attached to the thigh, shank, and foot segments are relevant to each segment motion. The reference frames (b) attached to the hip, knee, and ankle joints are relevant to each anatomical joint motion.

After a short period of warm-up, they were instructed to perform side-foot and instep kicks, using their preferred leg (right) with maximum effort, to the center of goal, which was located at a distance of 11 m in front of them. A regulation soccer ball (FIFA standard) was used. All subjects performed at least three attempts for each type of kick so that one shot could be selected with both a good foot–ball impact and adequate center of goal targeting. The quality of the impact was judged in each trial by both the participant and the present investigator. When the participants could not produce a successful shot within three attempts, kicks were repeated until an additional shot satisfied the above two criteria. Two electrically synchronized high-speed video cameras (NAC Inc., Tokyo, Japan) were used to sample the kicking motion at 200 fps (shutter speed was 1/2000 s), positioned at the rear and kicking leg (right) side. For calibrating the performance area, a calibration frame ($1.5 \times 1.5 \times 1.8$ m) with 16 control points was videotaped before the trails.

A digitizing system (DKH Inc., Tokyo, Japan) was used to digitize manually the anatomical body landmarks, including right and left shoulders, hips, knees, ankles, heels, and toes. The center of the ball was also digitized in its initial, stationary position and in all of the available frames after it left the foot (five frames). Each trail was digitized from the take-off of the right foot toe to the impact of the foot with ball. Ten additional frames (50 ms) before the right toe take-off and after the ball impact were also digitized to exclude the effect of distortion caused by filtering. The direct linear transformation method (1) was used to obtain the three-dimensional coordinates of each landmark. The three-dimensional coordinates were expressed as a right-handed orthogonal reference frame R_G, in which Z_G was vertical and pointed upward, Y_G was horizontal and pointed in the direction from the penalty spot to the center of the goal, and X_G was perpendicular to Y_G and Z_G. The performance area ($1.5 \times 1.5 \times 1.8$ m) was calibrated with a net root mean square error of 5 mm.

The horizontal component of the ball velocity was calculated as the first derivative of linear regression lines fitted to their nonfiltered displacements, and the vertical component of ball velocity was calculated as the first derivative of a quadratic regression line with its second derivative set equal to -9.81 m·s^{-2} fitted to its nonfiltered displacement in the available airborne frames. The absolute magnitude of the ball velocity was calculated from the value of its components.

For calculating absolute angular velocity and angular acceleration of each segment, local reference frames R_T, R_S, and R_F were defined at the center of thigh, shank, and foot segments, respectively (Fig. 1a). Z_T was a vector from the right hip to the right knee, Z_S was a vector from the right knee to the right ankle, and Z_F was a vector from the right toe to the right heel. X_T was the vector product of Z_S and Z_T, X_S was the vector product of Z_F and Z_S, and X_F was the vector product of a vector from the right heel to the right ankle and Z_F. Y_T, Y_S, and Y_F were the vector products of Z_T, Z_S, and Z_F and X_T, X_S, and X_F, respectively. All reference frame vectors were normalized to unit length.

According to the procedure described by Feltner and Dapena (4), the kicking leg was modeled as a three-link kinetic chain composed of the thigh, shank, and foot, in which the sum of all torques applied to each segment was set equal to the vector product of the segment's moment inertia and angular acceleration. Segment mass, center of the mass location, and moment of inertial values were derived from the data of young living Japanese athletes by Ae et al. (2) using a similar procedure first described by Jensen (7), which would be more appropriate for the participants in the present study. The mass of the shoe was 200 g, which was added to the foot mass, and its effect on the length of radius gyration of the foot segment was assumed to be small enough to be negligible. Using the methods similar to those reported by Sprigings et al. (15), the relative angular velocity vector of each joint was computed by subtracting its absolute angular velocity vector from that of the adjacent proximal segment. A reference frame was additionally defined at the center of trunk segment R_{TR} to compute the relative angular velocity vector of the thigh. Z_{TR} was a vector from the midpoint of a line segment between both hips to that between both shoulders, X_{TR} was a vector from the left hip to the right hip, and Y_{TR} was the vector product of Z_{TR} and X_{TR}. All reference frame vectors were normalized to unit length. Thus, the absolute angular velocity

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TABLE 1. Direction of anatomically relevant reference frames for torque and angular velocity.

	Positive (+)	Negative (−)
Hip joint (R_{AH})		
	Adduction	Abduction
	Flexion	Extenstion
	External rotation	Internal rotation
Knee joint (R_{AK})		
	Extension	Flexion
	External rotation	Internal rotation
Ankle joint (R_{AA})		
	Dorsiflexion	Plantarflexion
	Inversion	Eversion

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vector of the trunk was calculated and then subtracted from that of the thigh.

As shown in Figure 1b, both the torque and the relative angular velocity vectors were separated into orthogonal components using unit vectors included in anatomically relevant reference frames defined at hip (R_{AH}), knee (R_{AK}), and ankle (R_{AA}) joints. Z_{AH}, Z_{AK}, and Z_{AA} were equal to Z_T, Z_S, and $-Z_S$, respectively. X_{AH} was the vector product of a vector from the left hip to the right hip and Z_{AH}. Y_{AH} was the vector product of Z_{AH} and X_{AH}. X_{AK} was the vector product of Z_{AK} and Z_{AH}. Y_{AK} was the vector product of Z_{AH} and X_{AH}. X_{AA} was the vector product of Z_{AA} and Z_F. Y_{AA} was the vector product of Z_{AA} and X_{AA}. Hip torque and angular velocity vectors were separated into three components: adduction ($+X_{AH}$)/abduction ($-X_{AH}$), flexion ($+Y_{AH}$)/extension ($-Y_{AH}$), and external rotation ($+Z_{AH}$)/internal rotation ($-Z_{AH}$). Knee torque and angular velocity vectors were separated into two components: extension ($+X_{KA}$)/flexion ($-X_{KA}$) and external rotation ($+Z_{KA}$)/internal rotation ($-Z_{KA}$). Ankle torque and angular velocity vectors were separated into two components: dorsiflexion ($+X_{AA}$)/plantarflexion ($-X_{AA}$) and inversion ($+Y_{AA}$)/eversion ($-Y_{AA}$). The direction of the torques and angular velocities separated into the anatomical-relevant reference frame are referenced in Table 1.

The impact of the foot with the ball produces a sudden deceleration of the kicking leg, which causes a serious distortion of the kinetic data near impact. For avoiding such systematic errors, the time-dependent coordinate data were digitally smoothed in a forward order (toward the impact) by a second-order Butterworth low-pass filter (16) at 12.5 Hz. Although this smoothing procedure may have minimized the risk of systematic errors stemming from impact, the reverse-order filtering to eliminate phase distortion was canceled, thereby making the data prone to phase distortion.

As the joint torque data were a second derivative parameter, the instantaneous value can be influenced to a large extent by noise in the raw coordinate data (5). For minimizing potential interpretative errors, the joint torque was integrated using the trapezoidal rule for 50 ms. The integrated region shifted by a frame (5 ms) throughout the analyzed portion to find the region having the largest angular impulse. The peak magnitude of torque and time of its occurrence were computed as the average value and median time of the region (50 ms), respectively.

For simplifying interpretation of results, the kicking motion was divided into three phases by these four key moments (Fig. 2). The forward swing of the kicking leg was started from right toe off (RTO), with the hip joint reaching its maximum extension angle (MHE) while the knee flexion motion was occurring. The knee joint subsequently reached its maximum flexion angle (MKF) and then continuously extended until ball impact. The back-swing phase was defined as a time period from RTO to MHE. The leg-cocking phase followed and ended at MKF. The leg-acceleration phase subsequently occurred and ended at ball impact.

Comparisons were made between the two kicks using one-way paired Student *t*-tests. The criterion for statistical significance was $P < 0.05$ for all analyses.

RESULTS

Table 2 shows the initial ball velocity for each participant. The average ball velocities of the side-foot and instep kicks were 23.4 ± 1.7 $m \cdot s^{-1}$ and 28.0 ± 2.1 $m \cdot s^{-1}$, respectively. The initial ball velocity of the side-foot kick was significantly slower than that of the instep kick.

Table 3 shows the selected temporal variables. No marked differences were observed in temporal variables between the two kicks. As the total time (from RTO to ball impact) observed for the side-foot kick (190 ± 26.2 $m \cdot s^{-1}$) was shorter than that of the instep kick (221.0 ± 20.1 $m \cdot s^{-1}$), the time from RTO to ball impact was normalized to

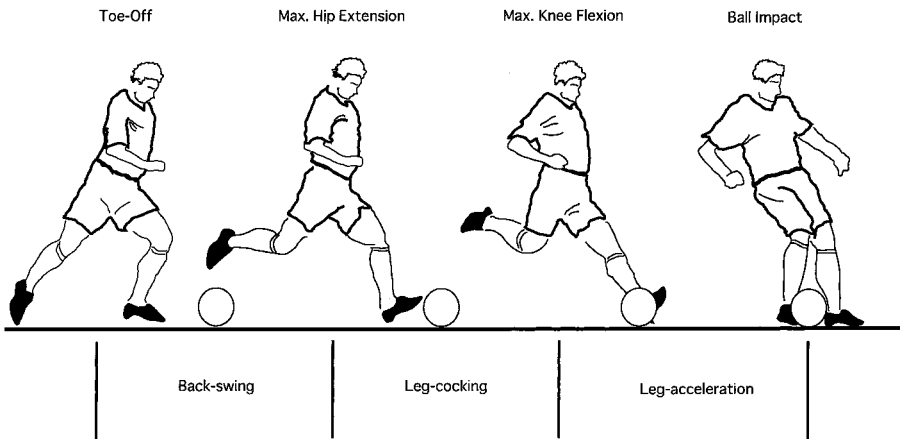

FIGURE 2—Definition of three phases of kicking.

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TABLE 2. Initial ball velocities ($m \cdot s^{-1}$).

Participant	Instep	Side-Foot	*P* Value
1	30.9	23.8	
2	27.5	25.4	
3	25.9	24.2	
4	26.3	22.5	
5	29.2	21.1	
Mean	28.0	23.4	0.013*
SD	2.1	1.7	

* Shows significant difference ($P < 0.05$).

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100%. The side-foot kick showed a smaller portion of the back-swing phase (33.7 ± 6.3%) and a larger portion of the leg-cocking phase (46.3 ± 5.2%) than those of the instep kick (48.5 ± 8.2% and 28.2 ± 8.8%, respectively).

The changes of average (±SD) joint torques and angular velocities of the hip, knee, and ankle are presented in Figures 3, 4, and 5, respectively. The peak magnitudes are summarized in Table 4. The general patterns of the hip flexion, hip adduction, and knee extension torques and angular velocities were similar in both kicks. Among the joint torques, the hip flexion torque was the largest in both kicks and their magnitudes were comparable between the two kicks (see Figure 3 and Table 4). In both kicks, the peak magnitude of the hip flexion torque occurred during the leg-cocking phase and the hip flexion angular velocity rapidly increased during the leg-cocking phase and reached similar peak magnitudes, after which the angular velocity was decreased until ball impact.

The hip adduction torque was recorded as the second largest torque in both kicks, and their magnitudes were also comparable between the two kicks (see Fig. 3 and Table 4). The hip adduction torque of the side-foot kick reached the peak magnitudes during the leg-cocking phase, whereas that of the instep kick varied among participants; the peak magnitudes occurred during the back-swing phase (one participant), leg-cocking phase (one participant), and leg-acceleration phase (three participants). Although slight hip adduction motion occurred close to ball impact in both kicks, their magnitudes were small enough to be negligible.

A remarkable difference was observed, however, for the hip external rotation torque and angular velocity between the two kicks (see Fig. 3). The hip external rotation torque was dominant only for the side-foot kick. The torque of the side-foot kick reached its peak magnitudes during the leg-cocking phase except for one participant, who reached it during the leg-acceleration phase. The magnitude of this torque of the side-foot kick was significantly larger than that of the instep kick (see Table 4). The hip external rotation angular velocity of the side-foot kick increased rapidly during the leg-acceleration phase, whereas that of the instep kick was very small. The magnitude of hip external rotation angular velocity was significantly larger than that of the side-foot kick (see Table 4).

The knee extension torque and angular velocity were dominant in both kicks, whereas those of the knee external/internal rotation torque were small enough to be negligible (see Fig. 4). The knee extension torque in each kick reached its peak magnitude before the knee extension motion was initiated (during the leg-cocking phase). The knee extension angular velocity in each kick subsequently increased very rapidly during the leg-acceleration phase and reached its peak magnitudes at ball impact. No marked differences were observed between the two kicks for the magnitudes of the knee extension torque and angular velocity.

At the ankle, although dorsiflexion and inversion torques were consistently exhibited in both kicks, the change of angular velocities of dorsiflexion/plantarflexion and inversion/eversion were very small (see Fig. 5). The magnitude of the dorsiflexion torque of the instep kick was larger than that of the side-foot kick, whereas there were no marked differences for the inversion torque between the two kicks.

DISCUSSION

Ball speed. Several studies have reported the average ball velocities of the instep kick or similar maximal soccer kicks. The average ball velocity of the instep kick measured in the present study was within the range of average values (22.2–30.0 $m \cdot s^{-1}$) reported previously (3,9,10,17). Only one study (9) reported the ball velocity of both the side-foot and instep kicks. Thus, the values of Levanon and Dapena (9) can be used for comparison. As they also reported the velocity of the center of mass of the foot, for an accurate comparison, the foot velocity was calculated from the coordinate data measured in the present study. The center of mass location of the foot was defined by toe and heel markers, and its velocity was computed as the first derivative of linear regression line fitted to its nonfiltered displacement data in five frames immediately before the impact. Although high school players were used in the present study, the ball and foot velocities of the side-foot (23.4 ± 1.7 $m \cdot s^{-1}$ and 19.1 ± 1.1 $m \cdot s^{-1}$, respectively) and instep (28.0 ± 2.1 $m \cdot s^{-1}$and 20.3 ± 1.0 $m \cdot s^{-1}$, respectively) kicks were similar to the values reported by Levanon and Dapena (9) for collegiate players performing the side-foot (22.5 ± 1.8 $m \cdot s^{-1}$ and 18.3 ± 1.8 $m \cdot s^{-1}$, respectively) and instep (28.6 ± 2.2 $m \cdot s^{-1}$ and 21.6 ± 2.0 $m \cdot s^{-1}$, respectively) kicks, respectively. Levanon and Dapena (9) concluded that the slower ball speed observed for the side-foot kick was not due to the quality of impact but due almost exclusively to the slower final speed of the foot in that kick. The results of the present study confirmed their results by showing the similar slower ball and foot velocities of the side-foot kick than those of the instep kick.

TABLE 3. Selected temporal variables.

	Instep Mean (SD)	Side-Foot Mean (SD)
Duration of phases (ms)		
Back-swing	108.0 (25.9)	65.0 (20.3)
Leg-cocking	62.0 (19.9)	87.0 (5.7)
Leg-acceleration	51.0 (7.4)	38.0 (5.7)
Total	221.0 (20.1)	190.0 (26.2)
% of total duration (%)		
Back-swing	48.5 (8.2)	33.7 (6.3)
Leg-cocking	28.2 (8.8)	46.3 (5.2)
Leg-acceleration	23.2 (3.9)	20.0 (1.8)

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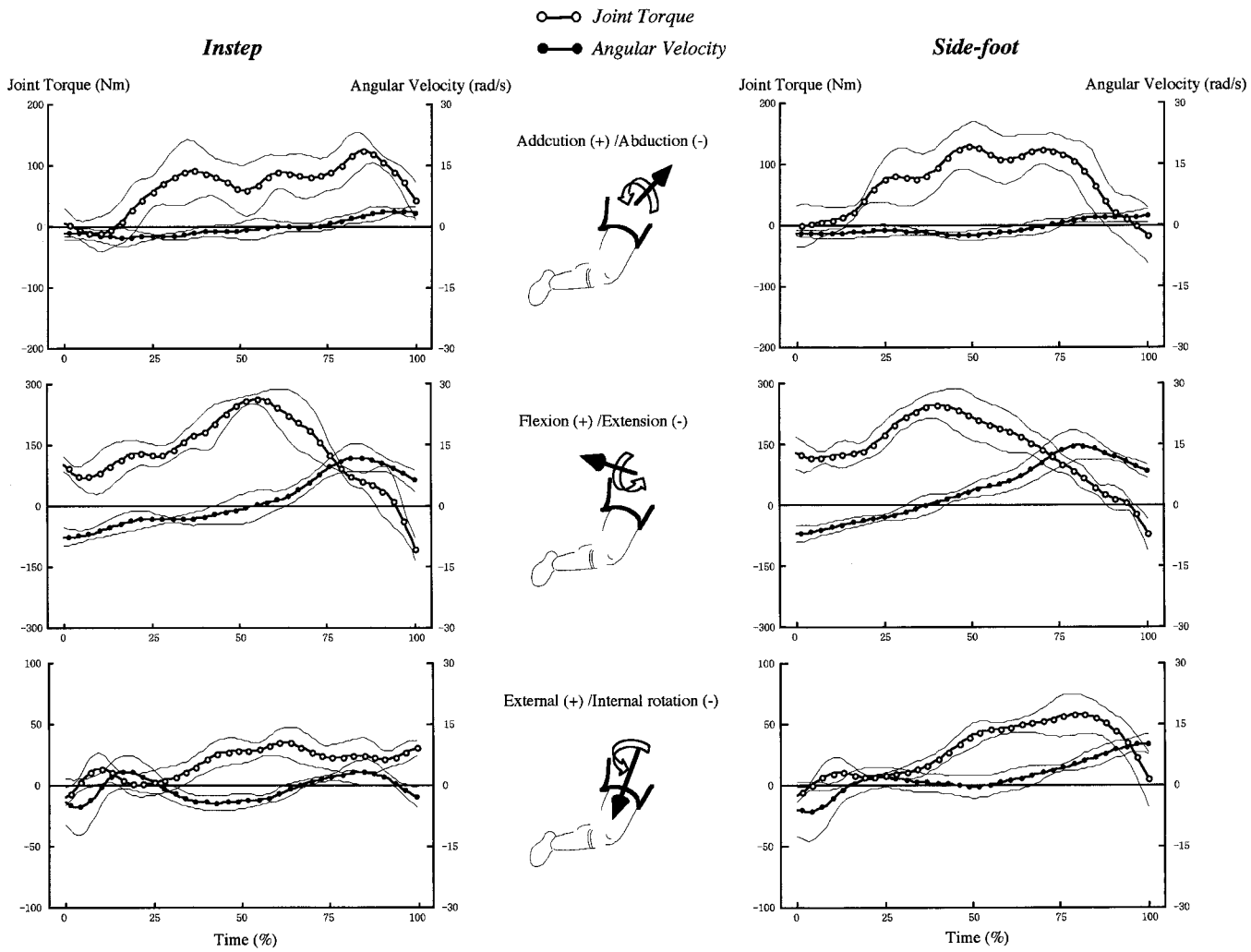

FIGURE 3—Changes in joint torque and angular velocity at the hip of the side-foot (right column) and instep (left column) kicks. For the hip adduction/abduction (top), positive and negative values corresponded to adduction and abduction, respectively. For the hip flexion/extension (middle), positive and negative values corresponded to flexion and extension, respectively. For the hip external/internal rotation (bottom), positive and negative values corresponded to external and internal rotation, respectively.

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Instep kick kinetics. Kicking biomechanics have been studied by several authors using two-dimensional procedures (3,10-13,17). Thus, among the kinetic parameters obtained in the present study, the values of hip flexion and knee extension torques were comparable to the literature. The previous studies regarding the instep or similar maximal soccer kicks consistently found that the hip joint generates the largest flexion torque, which is approximately twice as large as the knee extension torque. In the present study, for minimizing potential interpretative errors, the magnitudes of torques were computed as the average values. Putnam's (11) was the only study that reported the averaged magnitudes for the hip flexion and knee extension torques, and her values (hip flexion = 229 ± 34 N·m; knee extension = 85 ± 12 N·m) were comparable to those of the present study (hip flexion = 249 ± 31 N·m; knee extension = 98 ± 27 N·m). However, the other previous studies reported the instantaneous peak values, which seem to vary considerably among the literature (hip flexion = 194–274 N·m; knee extension = 83–122 N·m). Thus, the instantaneous values were reevaluated from the data of the present study for an equitable comparison. Although the instantaneous values of the present study (hip flexion = 283 ± 30 N·m; knee extension = 111 ± 39 N·m) were comparable to those reported by Zernicke and Roberts (17) (hip flexion = 274 ± 36 N·m; knee extension = 122 ± 23 N·m), those reported by Robertson and Mosher (13) (hip flexion = 220 N·m; knee extension = 90 N·m) and by Luhtahnen (10) (hip flexion = 194 ± 33 N·m; knee extension = 83 ± 21 N·m) were substantially smaller than those of the present study.

The coordinate data of the present study were smoothed only toward ball impact, whereas those in the previous studies were bidirectionally smoothed. The influence of the smoothing procedure conducted in the present study must be examined. For investigating this issue, the coordinate data of the present study were additionally smoothed in the reverse direction and reevaluated the averaged magnitudes of the hip flexion and knee extension torque. The values from bidirectionally smoothed data (hip flexion = 249 ± 31 N·m; knee extension = 100 ± 28 N·m) were similar to those

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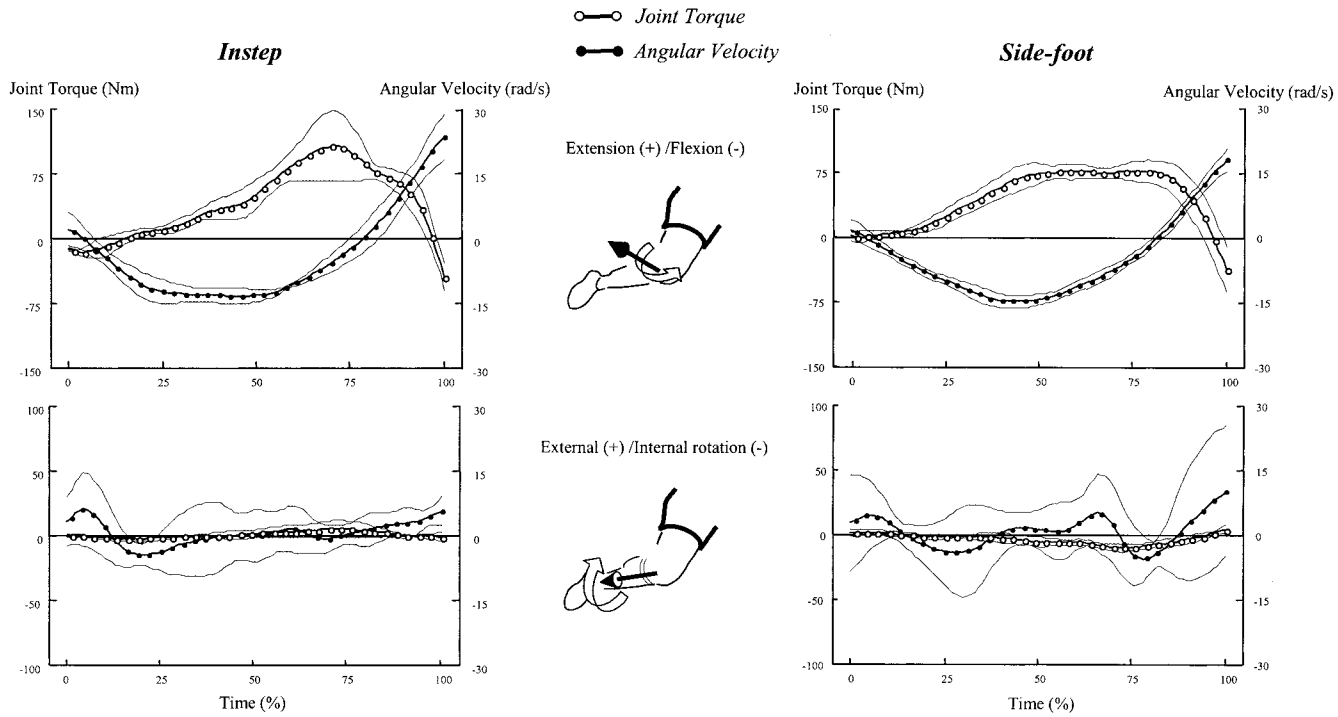

FIGURE 4—Changes in joint torque and angular velocity at the knee of the side-foot (right column) and instep (left column) kicks. For the knee extension/flexion (top), positive and negative values corresponded to extension and flexion, respectively. For the knee external/internal rotation (bottom), positive and negative values corresponded to external and internal rotation, respectively.

derived from unidirectionally smoothed data (hip flexion = 249 ± 31 N·m; knee extension = 98 ± 27 N·m). Thus, the smoothing procedure used in the present study had little effect on the magnitudes for the hip flexion and knee extension torques. Meanwhile, their occurrence times were shifted toward ball impact. The occurrence time of the peak hip flexion torque was systematically shifted for 15 ms (three frames) toward ball impact, whereas that of the knee

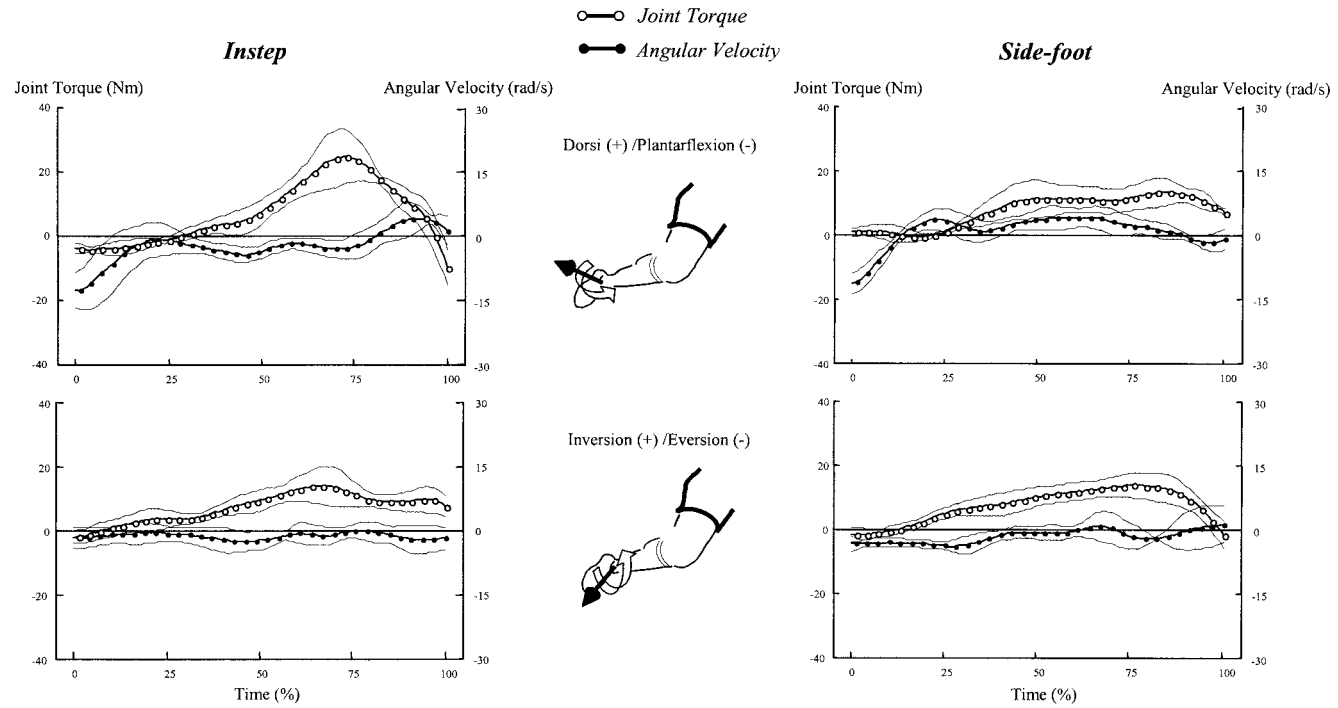

FIGURE 5—Changes in joint torque and angular velocity at the ankle of the side-foot (right column) and instep (left column) kicks. For the ankle dorsiflexion/plantarflexion (top), positive and negative values corresponded to dorsiflexion and plantarflexion, respectively. For the ankle inversion/eversion (bottom), positive and negative values corresponded to inversion and eversion, respectively.

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TABLE 4. Selected values of joint torque (N·m) and angular velocity (rad·s^{-1}).

	Instep Mean (SD)	Side-Foot Mean (SD)	*P* Value
Torques			
Maximal ankle dorsiflexion torque	22 (6)	13 (3)	
Maximal knee extension torque	98 (27)	80 (7)	
Maximal hip adduction torque	115 (17)	129 (23)	
Maximal flexion torque	249 (31)	231 (38)	
Maximal external rotation torque	33 (8)	56 (12)	0.002*
Angular velocities			
Maximal knee extension angular velocity	23.8 (5.2)	18.3 (2.5)	
Maximal hip flexion angular velocity	12.9 (3.0)	15.5 (3.3)	
Maximal hip external rotation angular velocity	6.0 (2.0)	11.1 (2.4)	0.017*

* Shows significant difference ($P < 0.05$).

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extension torque was not systematically shifted; the times ranged from 10 to 25 ms. Some kinetic parameters of the proximal segment are typically affected by a sudden deceleration produced by foot–ball impact, which may vary its shifted time.

Rodano and Tavana (14) conducted a three-dimensional analysis of the instep kick and found a larger discrepancy between the two- and three-dimensional analyses for angular measures. However, it is still unknown whether joint torque data calculated in the two-dimensional procedure were over- or underestimates compared with those from the three-dimensional procedure. One possible explanation for the considerable discrepancy for the magnitudes of hip and knee torques between the previous studies and the present study is the difference in filming procedure; data were collected three-dimensionally in the present study and two-dimensionally in the previous studies. For clarifying this issue, the torque vectors calculated at the hip and knee joints for the instep kick were projected onto the sagittal plane (the Y-Z plane of the reference frame R_G), and their magnitudes were reevaluated. The magnitude of the knee extension torque was slightly but consistently decreased when the vectors were projected onto the sagittal plane (98 ± 27 N·m vs 91 ± 21 N·m), whereas that of the hip flexion torque was not consistently affected (249 ± 31 N·m vs 245 ± 33 N·m). A more distal segment tended to have a larger out-of-plane motion that may cause an underestimation of the extension torque when the motion was projected onto the sagittal plane. Thus, the two-dimensional procedure conducted in the previous studies partially explains the difference of the magnitude of knee extension torque.

A three-dimensional kinetic analysis first conducted in the present study for the instep kick did demonstrate that the hip adduction torque was continuously generated throughout the kicking (see Fig. 3). Although this torque had the second largest magnitude (see Table 4), the adduction motion was not so noticeable during the kicking. The motion-dependent interactions produced by an angled approach run and successive pelvis rotation and outward leg swing may act to abduct the hip joint. Thus, the hip adduction torque probably acts to control the hip abduction angle, thereby playing a role to guide the whole kicking leg in correct orientation during kicking.

A proximal-to-distal sequence of segment motions has been recognized in swing motions, in which it has been suggested, as a general principle, that the negative angular acceleration of a proximal segment aids to increase the positive angular acceleration of an adjacent distal segment. However, there is some disagreement regarding a source of deceleration of the proximal segment. As a small magnitude of hip extension torque was recorded in several studies (10,13,17), there has been a tendency to attribute the decrease in thigh angular velocity to the actions of the hip extensors. However, Putnam (11) found a substantial influence of the motion-dependent interaction during the kicking and threw doubt on the role of the hip extension torque by showing that the decrease of thigh angular velocity occurred in the presence of a large hip flexion torque. In the present study, although the hip flexion torque rapidly decreased to a small magnitude of the hip extension torque immediately before ball impact like most previous studies, the deceleration of the proximal segment had been initiated before the hip extension torque was exhibited (see Fig. 3). This indicates that the hip extension torque was not the only source to decelerate the proximal segment during the instep kicking. There was likely to be a substantial effect of the motion-dependent interaction on the proximal segment motion. Among the several motion-dependent interactions, the knee joint torque acting reversely on the distal end of the proximal segment may account for this phenomenon. However, without additional analysis focused on the motion-dependent interaction during the instep kicking, the results of the present study are still insufficient to determine the critical source for decelerating the proximal segment.

Side-foot kick kinetics. For the kinetic aspects, no comparisons to the literature were possible for the side-foot kick. Thus, an attempt was made to understand the mechanics underlying the side-foot kick through a comparison with those of the instep kick. Evidently, the changes of the hip flexion, hip adduction, and knee extension torques and angular velocities were similar to those found for the instep kick. No remarkable differences were observed between the two kicks for motion within the thigh-shank plane, in which combined hip flexion and knee extension generate the high foot velocity. If the leg motions of the side-foot kick are restricted within this plane, then players are forced to have some unlikely posture of the lower leg and foot to achieve perpendicular orientation of the medial side of the foot to the desired ball direction. The present study did demonstrate differences between kicks in the generated hip external rotation torque. For the side-foot kick, this torque was dominantly exhibited from the leg-cocking to leg-acceleration phase (see Fig. 3), and its magnitude (56 ± 12 N·m) was significantly larger than that of the instep kick (33 ± 8 N·m). Moreover, the hip external rotation angular velocity was dominant during the leg-acceleration phase, and its magnitude (11.1 ± 2.4 rad·s^{-1}) was significantly larger than that of the instep kick (6.0 ± 2.0 rad·s^{-1}). These differences found for the side-foot kick would provide a better solution for how to swing the leg to achieve sufficient square orientation and forward velocity of the medial side of the foot.

Figure 6 shows the average (± SD) angles of the pelvis (defined as the vector from left to right hip) and thigh-shank

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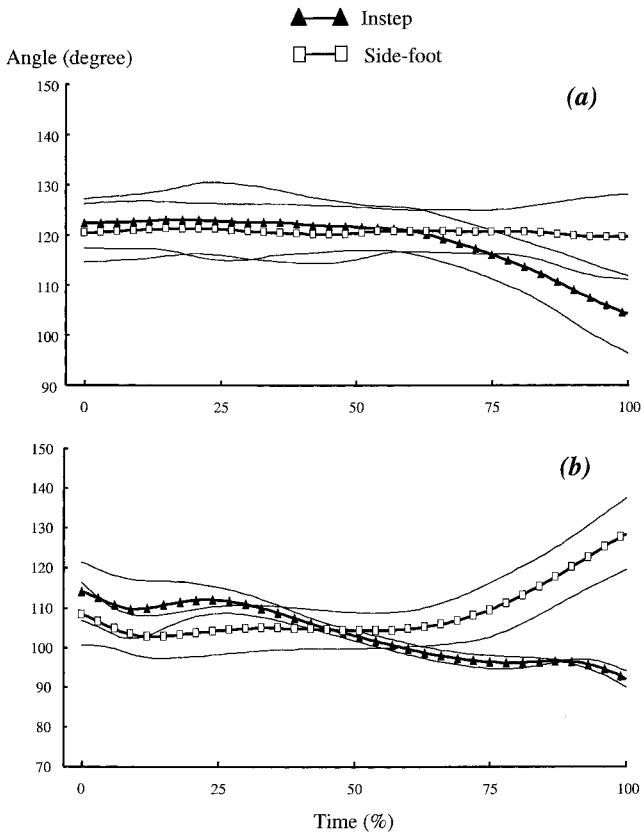

FIGURE 6—Average (±SD) horizontal angles of the pelvis (a) and thigh-shank plane (b) relative to the anterior direction (the Y axis fixed on the ground) of the side-foot and instep kicks.

plane (defined as the vector normal and pointing outward to the thigh-shank plane) relative to the anterior direction (the Y_G axis of the reference frame R_G) within the horizontal plane (the X-Y plane of the reference frame R_G). As shown, a clear difference was observed for the thigh-shank plane motion between the two kicks. In the side-foot kick, the thigh-shank plane rotated clockwise during the leg-acceleration phase. In the instep kick, by contrast, the plane rotated counterclockwise during the leg-cocking phase and reached mostly square orientation to the anterior direction. Although the instep kick also showed a counterclockwise rotation of the pelvis that of the side-foot kick was varied among the participants; two participants showed a counterclockwise rotation similar to that of the instep kick, one participant showed no appreciable rotation, and the other two participants showed a clockwise rotation opposed to that of the instep kick. Thus, the pelvis rotation was not clearly different between the two kicks. Hence, it is reasonable to suppose that the clockwise rotation of the thigh-shank plane observed for the side-foot kick was mainly due to the hip external rotation motion generated by the hip external rotation torque. This motion in the thigh-shank plane allowed players to face the medial side of the foot toward the direction that the ball should be launched.

The outward orientation of the thigh-shank plane was clearly observed for the side-foot kick during the leg-acceleration phase of kicking. This orientation, on the one hand, mostly restrains the contribution of the series of planar motions (the hip flexion and knee extension) to the forward foot velocity and, on the other hand, may allow the hip external rotation motion to increase the forward velocity of the foot. As shown in Figure 7, at the final phase of side-foot kicking, in the case of the thigh-shank plane pointing outward while the knee is not fully extended, the hip joint can utilize its external rotation motion to increase directly the forward velocity of the medial side of the foot. Consequently, because the available musculature used to derive this motion is typically smaller than that used for the planar motions, the foot velocity produced by this motion may be limited. This may account for the slower resultant foot velocity of the side-foot kick found in this investigation.

This aspect of the side-foot kick seems to be inconsistent with the results of angular motions reported by Levanon and Dapena (9). They reported that players (right footed) orient the pelvis, the right leg, and the foot more toward the right to hit the ball with the medial side of the foot. In their report, although the hip external rotation was recorded shortly before ball impact, the foot velocity was mostly produced by a series of planar motions: hip flexion and knee extension, as in the instep kick. From these results, they concluded that the hip external rotation did not contribute to speed of the foot but added a normal component of velocity to the thigh-shank plane. This makes it possible for the resultant velocity vector of the foot to point more perpendicular to the longitudinal axis of the foot, thereby facilitating the impact of the medial side of the foot squarely against the ball. The results of the present study, by contrast, strongly suggest a considerable contribution of the hip external rotation motion

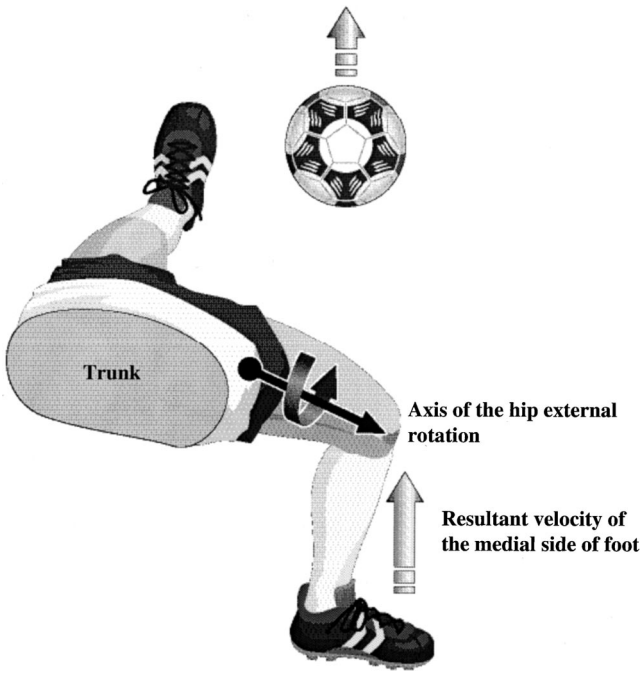

FIGURE 7—Overhead view of hypothetical mechanics of the side-foot kick. At the leg-acceleration phase, the thigh-shank plane rotated clockwise and reached its outward orientation. This allows the hip external rotation motion to increase directly the forward velocity of the medial side of the foot.

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to the forward velocity of the medial side of the foot. This apparent disparity may be exposed by a different style of the side-foot kick, which is used to produce a comparable ball velocity to the value reported by Levanon and Dapena (9).

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